Congpei Zhou

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Portfolio: https://bobhansky.github.io/portfolio/

Education

University of California San Diego Sep.2025 - Dec 2026

Computer Science (Master)

University of Minnesota Twin Cities Sep. 2021 - Aug. 2025

Computer Science (Bachelors) GPA *3.92/4*

Professional Experience

Apple Inc. (Cupertino, CA) Jun 2026 - Sept. 2026

Software Enginner Intern

- Media Processing. Video Rendering Pipeline. Video Playback. LLM Agent Development.

NetEase Games (Hangzhou, China) Sept. 2024 - Nov. 2024

Game Engine Development Engineer Intern

- Writing extension for **Python 3.12** (Cpython) with **C++ 17** to dump and load data with Msgpack format (**json-like**).
- Using shared resources mechanism to reduce memory usage when loading object from binary msgpack data to realtime game engine.
- Using **RAII** mechanism wrapper class for PyObject* to prevent memory leaks.

University of Minnesota (Minneapolis, MN) May 2023 - Sept. 2024

Research Assistant

- Building **Unity3D VR** projects, setting up gameplay logic (**C#**) with **Meta Oculus Quest2,** writing custom post-processing effects with built-in rendering pipeline. Debug experience in **Unity Frame Debugger, Unity OpenXR.**
- Experience with **HLSL shader** to create a fixed fovea restrictor with peripheral post-processing effect to mitigate the cybersickness using **VR** headset. Reducing the algorithm time complexity from O(N^2) to **O(2N)**

Projects (check my portfolio for more)

- **Monte Carlo Ray Tracing Renderer (C++, CMake, Git**) Jul. 2023 - Present

Github Link: bobhansky/TutuRenderer: A CPU based Renderer for personal interest and education. (github.com)

A CPU based offline **ray tracer** built for personal interest and education purpose. **C++** code completely written from scratch. The implemented state of the art integrator is **Bidirectional Path Tracing**. Features includes Microfacet Model, Multiple Importance Sampling, BVH, etc. Use **Multithreading (std::thread)** to facilitate rendering, speeding up the rendering by **12x**.

- **Vulkan Grass Rendering (C++, Vulkan)** Dec. 2025 - Dec 2025

Github Link: bobhansky/VulkanGrass: Vulkan real-time grass rendering

A real time grass field rendering demo. GPU programming. Used **C++**, **Vulkan**, and GLSL to set up Compute Pipeline, Render Pipeline, Descriptor sets etc for whole pipeline for grass physics animation, shadings. Used GPU side Culling (distance, orientation, frustum), tessellation (Level of Details) to boost the rendering time up to **4x** when drawing 16384 grass blades. A thorough Performance analysis is written in readme.

Skills

- **C++, Python, Rendering, Vulkan, OpenGL, GLSL/HLSL, CUDA, RenderDoc, Unity3D**
- Specialized in Computer Graphics, Rendering, GLSL, HLSL, Ray Tracing, real-time rendering performance bottleneck analysis and improvement.
- Knowledge in Machine Learning and Deep Learning.